

- 7 Overhead transmission lines are protected from lightning overvoltages by
 - (a)Counter poise wires
 - (b)Protector tubes
 - (c)Ground or shield wires above the main conductors
 - (d)Shunt reactors.
- 8 The purpose of insulation coordination is to
 - (a)Limit the overvoltages
 - (b)To protect the electrical apparatus against overvoltages
 - (c)To grade the insulation of different power apparatus and overhead lines such that the least important and easily replaceable apparatus flashes or fails first and the most important one is protected to the highest level.
 - (d)None of the above a, b or c.
- 9 Electrostatic voltmeters can measure
 - (a)Only dc voltage
 - (b)Both dc and ac voltages up to high frequency
 - (c)Impulse voltages
 - (d)Ac, dc and impulse voltages
- 10 In an impulse current generator the capacitors are connected in
 - (a)Series
 - (b)Parallel
 - (c)Connected in parallel while charging and in series while discharging
 - (d)Connected in series while charging and parallel while discharging
- 11 Impulse testing of transformers indicates
 - (a)Winding to ground insulation strength
 - (b)Winding to winding insulation strength
 - (c)Dielectric strength, quantity of insulation and processing
 - (d)Induced voltages in other windings during transients
- 12 In wet flashover tests, the conductivity of water used is
 - (a) $10 \pm 1.5 \mu$ Siemens
 - (b) $100 \pm 15 \mu$ Siemens at ambient temperature
 - (c) $45 \pm 10 \mu$ Siemens at room temperature
 - (d) $< 1.0 \mu$ Siemens at 27°C .

Q2. Solve the following.

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| A) Derive Poisson's Equation & also write Laplace Equation | 6 |
| B) Illustrate Finite Element Method of estimation of electric field Stress. | 6 |

Q3. Solve the following.

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| A) Derive the equation for Townsend's second coefficient of ionization in electronegative gases & also state the Townsend's criteria for breakdown | 6 |
| B) Write a brief note on Glow & Arc discharge in case of corona discharge. | 6 |

Q4. Solve Any Two of the following.

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| A) Explain Suspended Particle Theory , Cavitation & Bubble Theory in case of liquid insulating material. | 6 |
| B) Explain Thermal Breakdown & Electromechanical Breakdown in solid dielectric material. | 6 |

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| C) | Write various Insulating materials used in different parts of following electric equipment | 6 |
| a) | Power Transformers | |
| b) | Rotating Machines | |
| c) | Circuit Breakers & Cables | |
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| Q5. | Solve Any Two of the following. | |
| A) | Explain Simpson's theory of charge formation in cloud with diagram. | 6 |
| B) | Explain the term surge voltage & also state causes of surge voltage. | 6 |
| C) | Explain the measurement of high ac voltage using capacitive voltage transformer. | 6 |
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| Q6. | Solve Any Two of the following. | |
| A) | Explain the generation of high voltage using multistage Marx circuit. | 6 |
| B) | Explain the testing of pin type insulators & bushing. | 6 |
| C) | Explain various tests required to be performed on high-tension cables. | 6 |

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